

General Information:

introduction to bacterial transcription:

- [Website](#)
- Refer to [vocab page](#)

General info I should know:

- Resistance is an organism-level phenotype that is causally attributed to specific genetic variations
- The **H37Rv** reference genome is the standard reference genome for Mycobacterium tuberculosis
- The MTBC is a clonal group of species that is classified into 10 major genetically distinct populations (i.e., lineages) and 9 animal-adapted lineages.

Papers:

Foundational Papers:

[Green et al 2022:](#)

- A convolutional neural network highlights mutations relevant to antimicrobial resistance in Mycobacterium tuberculosis
- Anna original CNN for antibiotic resistance prediction, motivated the next 2 papers
- Trained two CNN models to predict either multi-drug or single-drug antibiotic resistance
- Found their models to have competitive AUC to SOTA with the note of multi-drug being more sensitive and single-drug being more specific.
- *These insights and choices of model allow for saliency analysis which shows new regions of interest not previously identified.*
- Showed how many sites we have no idea what happens when we mutate them

[Tasmin & Mohanty et al 2026](#)

- BIG-TB: A benchmark for prediction and interpretability of sequence-based machine learning using Mycobacterium tuberculosis genomes
- Main authors: Mahbuba Tasmin and Saishradha Mohanty
- Benchmark paper for prediction from genome tuberculosis
- 17,000 short read sequences and antibiotic resistance measures
 - Also curated list of resistance conferring variants
- *Two tasks*
 - *Prediction of antibiotic resistance (main task for my project)*
 - *Attribution of prediction to known resistance-conferring variants*
- Shows SOTA performance from simpler CNN model
- Useful Findings:

- *On protein foundation models, representations that average over sequence positions perform poorly on both tasks*
- *Detailed breakdown of model performance on*
 - *DNA based beats protein based, hypothesis is that protein based misses non-coding regions*
 - *CNN beats all other models on DNA sequences particularly for single-drug*
 - *For protein-based CNN models still topped chart but less clearly*
- Models:
 - Simple ML on reference-alternate encodings
 - CNN and transformer on 1-hot
 - CNN on embeddings from ESM and DNA-BERT

Kulkarni et al 2026:

- Convolutional neural networks quantify antibiotic resistance in Mycobacterium tuberculosis with diagnostic grade accuracy and predict treatment response
- predicted quantitative phenotypes instead of binary phenotypes as before; fused amino acid features
 - They use CNNs to predict minimum inhibitory concentrations (MICs) for 8 antibiotics from MTBC (TB) gene sequences
- Found that low level drug resistance (even well below CC) increases full resistance development during treatment
 - This is generally only measured MICs which is not done in clinical practice
- They introduce SOTA CNN models to predict MICs from whole gene sequences (WGS)
- **What they added:**
 - **Adding amino-acid biophysical properties and lineage information to model** inputs improves accuracy over nucleotide-only models.
 - we added molecular weight (g/mol), isoelectric point, and hydrophobicity (Eisenberg scale) for each amino acid in the proteins encoded by the relevant genes and passed them through independent convolutions
 - molecular weight (g/mol), isoelectric point, and hydrophobicity (Eisenberg scale) – as a 3 x K matrix, where K is the length of a given protein.
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- Be careful when adding modalities, adding lineage decreased performance for one drug due to small dataset size and “strong lineage skew”
- In all 8 drugs the CNN achieves higher performance than simpler regression

Good Concept poor execution, main competition:

Scalable de novo classification of antibiotic resistance of Mycobacterium tuberculosis

- <https://pubmed.ncbi.nlm.nih.gov/38940175/>
- Designed a classifier using linear and non-linear models
 - LASSO logistic regression and Random Forests

- To train our model, we assemble the sequence data into genomic contigs, identify all unique 31-mers in the set of contigs, and build a feature matrix M , where $M[i, j]$ is equal to the number of times the i th 31-mer occurs in the j th genome.
 - This results in over 350 million 31-mers so they use a sparse matrix
- Mykrobe (Bradley et al. 2015, Hunt et al. 2019), which was initially released in 2015 and then updated in 2019, relies on the construction and analysis of a de Bruijn graph from catalogues of resistant and susceptible alleles on different genetic backgrounds, along with a set of antibiotic resistance genes. Together, they form what the authors refer to as a reference graph.
- To the best of our knowledge, this is the first machine-learning method that is trained in a de novo fashion, meaning that it only uses the oligonucleotides and AST data for training;

Leveraging large language models to predict antibiotic resistance in *Mycobacterium tuberculosis*

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC12261485/>
- We identified (annotated) all functional sequences and genes from the 12 185 assemblies using Prokka (Seemann 2014), which integrates ab initio gene prediction, sequence similarity searches, and curated databases to deliver comprehensive and accurate annotations.
- We developed a post-processing method to merge partial gene alignments into full-length sequences.
- Useful:
 - *To enhance the model's ability to recognize gene regulatory elements, we included intergenic regions flanking each of the genes of interest, capturing up to 350-bp upstream and downstream, with a minimum length of 50 bp.*
 - *Seems to have a fairly minor impact*

Genomic Diagnostics for Drug-Resistant *Mycobacterium tuberculosis*: Computational Prediction of Antimicrobial Resistance

- <https://www.biorxiv.org/content/10.64898/2026.05.25.727578v1.full.pdf>
- WGS-based prediction can return resistance calls in seconds to minutes per genome once sequencing is complete, vs. the 6–8 week wait for culture-based AST.
- Catalog-based methods impose minimal computational demands at deployment but miss mechanisms absent from the reference list, with a structural delay between variant discovery and catalog incorporation. De novo methods can capture multivariate and noncanonical signals, but MTB++ training requires hundreds of gigabytes of memory, and CRyPTIC-scale analyses require multi-terabyte memory across multiple nodes.
- Because ML performance is sensitive to training-set size, with predictive performance rising sharply once resistant isolates per drug exceed approximately 500, even modest expansions of the resistant-isolate pool for newer/repurposed drugs could yield substantial gains.
- MTB++ matched or exceeded LLMTB on the antibiotics tested — on rifampicin, MTB++ achieved 89.86% accuracy vs. LLMTB's 90.94%; on isoniazid, MTB++ reached 94.90% vs. LLMTB's 93.61%. Once feature selection identifies informative k-mers, random forest and logistic regression match or exceed transformer-based architectures at a fraction of the computational cost.

Datasets:

MycoExpress:

- <https://mycoexpress.wpi.edu/>

Genome Browser:

- <https://mycobrowser.epfl.ch/genes/Rv0262c>
- Suggestion: find the source of the data in the left panel, which allows you to turn on different visualization tracks about gene expression and essentiality.

Summary:

In short, simpler methods tend to work better, though not too simple. CNN seems to be the sweet spot with effective performance without overly complex models.

Model Type	Modality	Task	Performance	Claims
BERT-based LLM LLMTB	DNA sequence	Single-drug	First line F1-scores > 0.94 (usually) Second line lower, e.g. 0.086	good cross validation for first line drugs. Much worse for second line drugs
LogReg Big-TB	DNA sequence (ref-alt encoding)	Single-drug	AUC 0.865	Baseline method Competitive with complex models
CNN	DNA sequence (1-hot encoded)	Single-drug	AUC 0.8753	-
CNN	DNA sequence (1-hot encoded)	multi-drug	AUC 0.8875	Best model in set
CNN	DNABERT embed-full	single-drug	AUC 0.8460	
CNN	DNABERT embed-PCA10	single-drug	AUC 0.8392	
LogReg	Protein sequence (ref-alt)	single-drug	AUC 0.7378	In general DNA models out perform protein models
CNN	Protein sequence (1-hot encoding)	single-drug	AUC 0.7889	Best protein model
transformer	Protein sequence (1-hot encoding)	single-drug	AUC 0.6379	
ESM-320		single-drug	AUC 0.7651	
ESM-PCA10		single-drug	AUC 0.7871	
CNN Green et al. (2022)	DNA sequence, aligned 18 loci 1-hot encoding	multi-drug	AUC 0.82-0.995	effectively maps out and exploits correlation of

				multi-drug development
CNN	DNA sequence, aligned loci (only drug relative loci) 1-hot encoding.	single-drug	AUC 0.801-0.971	Ignores confounding non-causal variants
CNN Kulkarni et al. (2026)	DNA + protein + metadata. Fused WGS alignments, amino acid traits, and lineages	MICs prediction for 8 antibiotics	89% "essential - agreement" accuracy	Biophysical arrays successfully forced the network to identify the severe functional damage of nonsense truncations vs missense changes.
Ridge Linear Regression	DNA + protein. Flattened vectors	MICs prediction	-	Baseline method consistently beaten by above CNN
LASSO + RF Serajian et al. (2024)	DNA sequence. Sparse matrix from 31-mers	De novo multi-drug	F1 > 0.8 on first line F1 < 0.33 on second line	Allows for ultra-light weight deployment but weak performance
GNN Nguyen et al. (2026)	Multi-Representation DNA Features. Reference-free unitigs; SNPs and FCGR distance matrices (edges).	Multi-drug / multi-pathogen	Mean AUROC > 0.9	The clonal decoupling mechanism significantly boosted diagnostic scores by removing lineage-linked confounders.